Profit and Profit Maximisation for Monopolies

LLE Mathematics and Statistics

1. You are given the demand function and the total cost function for a monopolist. Use this information to calculate the profit function, π, and the price, quantity, and profit that maximises the profit.

   (a) Market demand: $Q = 100 - P$

      i. $TC = 250 + 20Q$

      ii. $TC = Q^2 - 4Q + 200$

      iii. $TC = 100 + 2\ln(Q + 1)$

   (b) Market Demand: $Q = 500e^{-0.2P}$

      i. $TC = 100$

      ii. $TC = 10 + 15Q$

2. A fish a chip shop knows demand for cod and chips is different for pensioners compared to those who are not pensioners.
   Demand for pensioners: $Q_P = 200 - 4P_P$ where P_P is the price charged to pensioners.
   Demand for others: $Q_N = 160 - 2P_N$ where P_N is the price charged to non-pensioners.
   The shop has fixed costs of 100 and variable costs of $0.5Q$

   Find the maximum profits if:

   (a) The shop charges the same price to both groups, $P_P = P_N$

   (b) The shop charges different prices to both groups.